



The Role Of IoT & AI In Battery Management Of Electric Vehicles

A good battery gives an electric vehicle the power to traverse long distances without any tension. But when upgraded with next-gen technology such as IoT and AI the the entire system is bound to become even more enhanced

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Battery management system (BMS) is a crucial part of an electric or autonomous vehicle. But what has the Internet of Things (IoT) or artificial intelligence (AI) got to do with the battery? Let's see.

The term IoT, coined in 1999, refers to the interconnection of sensors and software on a network. It helps control the connected devices, and their relevant data can be collected and visualised with the help of a user interface.

The main functions of IoT are:

1. Collect. Gather data from sensors and various other devices in a home, vehi-

cle, or manufacturing plant.

2. Communication. Send the data to a destination such as cloud, AWS, Azure, or a personal data centre using a communication gateway through Wi-Fi, 4G, or 5G, etc.

3. Analysis. Collect useful information from the raw data to visualise, generate reports, and to make predictions.

4. Action. Based on the analysis, communicate with other devices through machine-to-machine (M2M) protocols, or send notifications (SMS, email, etc) to the concerned persons for rectifying any errors.

Most people think that cars such as Tesla's Model S only can be classified as EVs. But that's not the case. We have had EVs in India for a long time, such as electric trains which are also EVs. Their engines are traction motor based. The central part of an EV can be a traction motor, a BLDC motor, or an AC induction motor.

EVs are catching attention because of their advantages such as:

- Much lesser dependency on petrol, diesel, or other fossil fuels
- Ensuring environment is free from pollution and noise
- Come with recyclable lithium-ion (Li-ion) batteries
- Low maintenance and low operation cost

The essential components of an EV include (see Fig. 2):

Motor. It can be an AC induction motor or a brushless DC (BLDC) motor. Although there is a massive debate raging on which

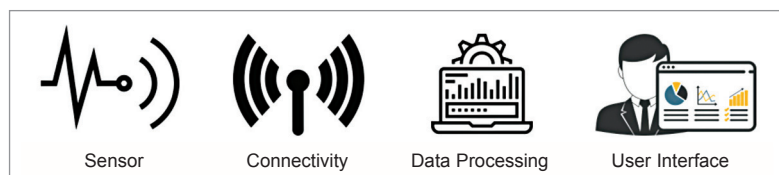


Fig. 1: The main components of IoT

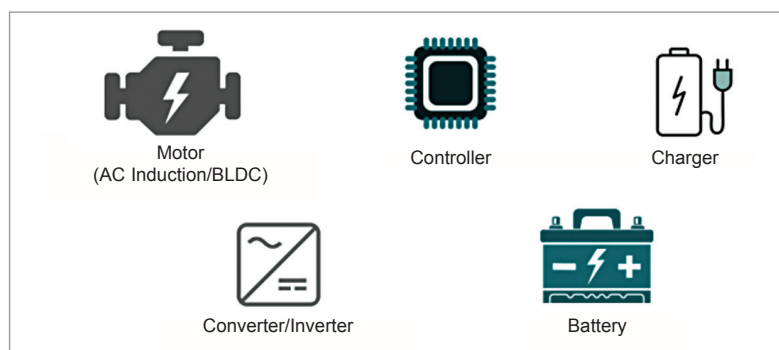


Fig. 2: Essential components of an EV